

Luiz Carlos Costa da Silva

AI Backend Engineer

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Portuguese (Native) • English (Advanced) • Mandarin (Learning)

Summary

AI Backend Engineer focused on production AI systems and Python backends. At Auria AI, I build and maintain **20+ production agents** supporting **3,000+ daily B2B security incidents**, maintain a PostgreSQL/SQLAlchemy data layer across **36 production tables**, and develop monitoring and alerting tools. Previously worked on a large Ruby on Rails to Kotlin migration at IME-USP. B.Sc. in Computer Science from the University of São Paulo.

Skills

Backend: Python, FastAPI, asyncio, Pydantic, SQLAlchemy, REST APIs, Microservices, Pytest
AI / LLMs: AI Agents, RAG, Embeddings, Prompt Engineering, Structured Outputs, LangChain, Pydantic-AI, Gemini, OpenAI API, Claude
Data & Tools: PostgreSQL, SQLite, Redis, Docker, Git, Metabase, Twilio API, Vector Search, CI/CD
Frontend: TypeScript, JavaScript, Vue.js, React, HTML, CSS

Experience

Backend Software Engineer August 2025 – Present
Auria AI São Paulo, Brazil

- Build and maintain Python AI agents across the production incident-response pipeline, supporting **3,000+ daily B2B security incidents** through summarization, insight extraction, and automated decision-making across **20+ agents**.
- Bootstrapped and maintain the PostgreSQL/SQLAlchemy data layer across **36 production tables**, shipping backend features end-to-end across multiple FastAPI services.
- Built **Auria's Metabase BI from scratch**, including operational dashboards, KPI analyses, and threshold-based alerts used daily by incident-response leadership.
- Designed and shipped an **on-call alerting bot** that calls responsible engineers through Twilio with AI-generated incident summaries, reducing overnight outage detection from **8+ hours to near-immediate**.

Software Developer (Research Scholarship) January 2021 – August 2023
Hub USP Inovação – IME-USP São Paulo, Brazil

- Migrated a large legacy Ruby on Rails system to a Kotlin backend, redesigning domain models, REST APIs, automated tests, and Docker deployment infrastructure.
- Maintained AUSPIN's innovation platform, serving **1,000+ patents**, **3,000+ companies**, **3,000+ professors**, and **100+ innovation initiatives**.

Projects

Memoir – AI Meeting Knowledge Workspace 2026 – Present

- Built an open-source application that turns meeting transcripts into summaries, decisions, action items, participants, and structured transcript segments.
- Implemented automatic format detection and parsers for Zoom and Google Meet, with a Gemini fallback for messy or unsupported transcript formats.
- Built the application with FastAPI, Pydantic-AI, SQLAlchemy, SQLite, vanilla JavaScript, HTML, and CSS, with Pytest coverage for parsing and core flows.

Divination – RAG Assistant for D&D Dungeon Masters 2024

- Built a RAG assistant grounded in the official 2024 D&D Free Rules using LangChain, vector search, and GPT-4o.
- Implemented a FastAPI backend and Vue 3 frontend as my B.Sc. capstone project at IME-USP.

Education

B.Sc. in Computer Science, University of São Paulo (IME-USP) 2020 – 2024

Leadership & Activities

President & Member, USPCoDeLab January 2021 – March 2024
IME-USP São Paulo, Brazil

- Served as President from January 2022 to August 2023, leading **20+ members**; organized two university hackathons, secured sponsorships, mentored participants, and managed event budgets.